

Clarification in Conversational RAG: Experiments, Pitfalls, and a Path Forward

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October 19, 2025

Abstract

We attempted to add “ask-one” clarifying questions to a production Retrieval-Augmented Generation (RAG) assistant without harming latency or groundedness. We explored two families of solutions: **(A)** an LLM-driven clarification policy embedded in the generation prompt; and **(B)** a deterministic retrieval-dispersion gate using entropy/perplexity over top results, with template-based clarifying questions. This report documents what we built, what failed on real conversations (e.g., “*show all bank of america products*”), why it failed, and what conditions must hold for each approach to become viable. We conclude with a pragmatic next step: postpone clarification and introduce an intent classifier that governs RAG behavior (including out-of-domain and vague utterances) and can route to intent-scoped indices or fall back to the full collection when warranted.

1 System Context and Baseline

Baseline. The assistant is a RAG system with sentence-level chunking and dense embedding retrieval. The language model (LLM) answers strictly from provided grounding context, and abstains when context is insufficient. The LangGraph pipeline is `START` → `ingest` → `frustration` → `build_query` → `retrieve` → `answer` → `decide` → `finalize` → `END`. Prior to our experiments, clarification was only triggered by explicit user frustration (keywords) or if the LLM itself returned a clarifying question along with `abstained=true`. In practice, this rarely occurred.

2 Attempt A: LLM-Driven Clarification Policy

2.1 Design

We expanded the main RAG prompt to include a conservative clarification policy: the LLM should propose a clarifying question if a faithful answer would exceed a word budget (150 words) and cover multiple “facets”; otherwise produce a concise answer. The JSON schema included fields such as `should_clarify`, `estimated_answer_words`, and `facet_count`. The `decide` node then routed to clarification when `abstained=true && clarifying_question != ""`.

2.2 Observed Behavior and Failures

- **The model almost never asked to clarify.** Given sufficient context, the LLM prefers to answer rather than ask, due to helpfulness bias in instruction-tuned models.
- **Coupling to “abstain.”** Our router clarified only when the LLM abstained; however, ambiguity (wide scope) is *not* the same as insufficiency. The LLM rarely abstained on broad queries, so clarification rarely triggered.

- **Length planning is weak.** Asking the LLM to predict word counts or number of facets before generation was poorly calibrated; once decoding starts, models tend to complete an answer rather than pivot into a question.

2.3 Example

User: “explain all bank of america products” *Actual:* Often a long answer or abstention; rarely a clarifying question. *Why:* The LLM had enough context to attempt an answer, so it did not abstain and our clarify path remained dormant.

2.4 Viability Conditions

For an LLM-only policy to work reliably, the system must enforce *structured gating* in post-processing: reject over-length, multi-facet answers and convert them into a single clarifying question. This adds complexity and erodes latency gains. Without such enforcement, precision of the clarify decision is not high enough for production.

3 Attempt B: Deterministic Clarification Gate (Entropy/Perplexity)

3.1 Design Overview

We moved the clarify decision outside the LLM and measured dispersion over the top- K retrieved *documents*. Let r be 1-based rank and $w_r = 1/\log_2(2+r)$. For document d , with ranks $\mathcal{R}(d)$ in the top- K :

$$p(d) = \frac{\sum_{r \in \mathcal{R}(d)} w_r}{\sum_{r=1}^K w_r}, \quad H_{\text{doc}} = - \sum_d p(d) \ln p(d), \quad E_{\text{doc}} = \exp(H_{\text{doc}}). \quad (1)$$

We interpret E_{doc} as the *effective number of distinct document modes*. We then maintain a tiny online scalar μ (words per mode) and predict faithful answer length as $\hat{L} = \mu \cdot E_{\text{doc}}$. The gate asks exactly one clarifying question iff: retrieval is sufficient, $E_{\text{doc}} \geq 3$, top-mass $p_{\text{max}} < 0.55$, and $\hat{L} > 150$ words. The clarifying question is template-based using top document labels; optional LLM rephrase is allowed under strict constraints (same options, JSON-validated).

3.2 Labeling and Question Text

Labels are derived from per-document metadata: prefer **title** $\rightarrow$ **section** $\rightarrow$ cleaned **source** path. The question template is:

This can be concise or comprehensive. Prefer a full overview of everything, or focus on one of: $\{label_1, label_2, label_3\}$?

We always include chips for **Full overview** and **Other**.

3.3 What Failed in Our Runs

- **Gate computed on too few hits ($K = 5$).** With only five results, E_{doc} rarely exceeded 3 even for broad queries. Our logs showed $E_{\text{doc}} \approx 1.9$ and $p_{\text{max}} \approx 0.65$ —not enough to trigger the gate.
- **Dirty labels.** Boilerplate like “skip to content” and cookie banners polluted labels, preventing meaningful options.
- **Calibrator drift.** We inadvertently updated μ even when the LLM abstained, biasing predicted lengths.

- **No post-LLM fallback.** When the LLM abstained on a broad query, we still returned abstain instead of asking the deterministic clarifying question.

3.4 Examples with Logs (Fits Page Width)

Query	Expected	Gate Inputs	Actual	Root Cause
what is tier	Answer	$E_{\text{doc}}=1.88, p_{\text{max}}=0.68$	Answer	Correct: single-mode, concise
show all bank of america products	Clarify	$E_{\text{doc}}=1.92, p_{\text{max}}=0.65$ (with $K=5$)	Abstain	K too small; label noise; no fallback

3.5 Viability Conditions

This deterministic gate is viable if and only if the following hold:

1. **Evidence depth:** compute dispersion on $K \in [30, 50]$ (metadata-only retrieval is fine); still send a small k to the LLM for context.
2. **Stable parent IDs:** chunks must carry a canonical `parent_doc_id`; dedupe near-duplicates before computing $p(d)$.
3. **Label hygiene:** sanitize labels against boilerplate (e.g., remove “skip to content”, images, cookie text).
4. **Update μ only on delivered answers.** Never on abstain/clarify turns.
5. **Post-LLM fallback:** if LLM abstains, trigger the deterministic CQ when E_{doc} indicates ambiguity.

Without these, the gate will under-trigger (low recall) and UX regresses to abstain.

4 Why Clarification is Hard for LLMs

- **Helpfulness bias:** instruction-tuned models tend to answer rather than ask, if enough context exists to assemble any answer.
- **Ambiguity $\neq$ insufficiency:** Clarification is a scope control problem; abstain policies capture missing facts, not breadth.
- **Planning vs decoding:** Reliable length/facet forecasting would require pre-planning; single-pass generation commits early and rarely pivots to questions.
- **Ambiguity has many forms:** referential (pronouns), taxonomic breadth (“all products”), underspecified constraints (“best account”), and temporal scope—a single general rule is hard to learn implicitly.
- **Retrieval masks ambiguity:** rich context makes questions appear answerable; the model follows the path of least resistance.

5 Decision and Next Step: Intent Classifier First

Given maintenance cost and the strict preconditions above, we defer clarification and adopt an **intent classifier** to govern RAG behavior.

5.1 Plan

1. **Taxonomy:** define a compact set of intents covering the corpus (e.g., Checking, Savings, Cards, Loans, Preferred Rewards, Fees, Eligibility, Overview/All, OOD).
2. **Training data:** bootstrap with silver labels from logs (query $\rightarrow$ clicked doc family); optionally expand with LLM-assisted labeling and manual review for borderline cases.
3. **Routing:** if intent confidence $\geq \tau$: retrieve from the corresponding intent collection/index; else fall back to full collection.
4. **Vague/OOD:** for low confidence or OOD, show suggestions/chips (intent options) or abstain gracefully; allow “Overview/All” to explicitly opt into full-collection summaries.
5. **Fail-safe:** user can always click “Full overview” or reformulate; the classifier never blocks answering—it only scopes retrieval.

5.2 Benefits

- **Stability:** deterministic routing improves precision without relying on model self-assessment.
- **Maintainability:** intent definitions evolve with the corpus; no deep coupling to generation prompts.
- **Dual use:** the classifier aids both *query understanding* (govern behavior) and *index selection* (intent-scoped retrieval).

6 Conclusion

Our LLM-driven policy under-triggered (models prefer answering), and our first deterministic gate under-triggered due to insufficient evidence ($K=5$), noisy labels, and calibrator updates on abstains. Making clarification work well requires deeper retrieval instrumentation and careful hygiene that add maintenance burden. As a pragmatic next step, we will deploy an intent classifier that both governs RAG behavior (OOD, vague, and broad prompts) and enables intent-scoped retrieval—with a safe fallback to the full collection when confidence is low or the user explicitly requests an overview.